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Surgical Neuropathology Report

M/26/000346	Satviki, Patidar	*27.04.2018	

Entry: 08.06.2026 **Exit:** 18.06.26 - / FKA-FKA

Molecular-neuropathological Examination

DAkkS
Deutsche Akkreditierungsstelle
D-IS-13060-02-00

Analysis
DNA methylation analysis

Clinical data

Supplier:	Apollo Proton Cancer Centre, India
Supplier's diagnosis:	Diffusely infiltrating glioma, NOS
Localisation:	Left medial temporal lobe
Manifestation:	/
Request for	DNA methylation analysis

Material
We received 1 paraffin block with the external ID 1270/26 B, corresponding to our internal ID M/2026/346. H&E staining was used to select a suitable area for DNA extraction from the paraffin block. Tumor cell content 85%. Internal Tumor DNA ID: 460278.

Method
Analysis using Illumina Human Methylation EPIC array and internal classifier V12.8.

Findings

SentrixID:	210258770030_R02C01
Copy-Number Variation	Flat
MGMT-Promoter Status	Unmethylated

UKHD

Use of these results for scientific or other purposes only with explicit authorization of the Medical Director

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Methylation Classification Brain v12.8:

Predictions	Calibrated score	Interpretation	
Low-grade glial/glioneuronal/neuroepithelial tumours	0.99774	match	✓
pilocytic astrocytoma	0.99596	match	✓
supratentorial midline pilocytic astrocytoma	0.9793	match	✓
MC Pilocytic astrocytoma, midline	0.9793	match	✓

Legend (brain)*: ✓ Match (score >= 0.9) ✗ No match (score < 0.9): possibly still relevant for cases with low tumor content and poor DNA quality.

Hierarchy levels:

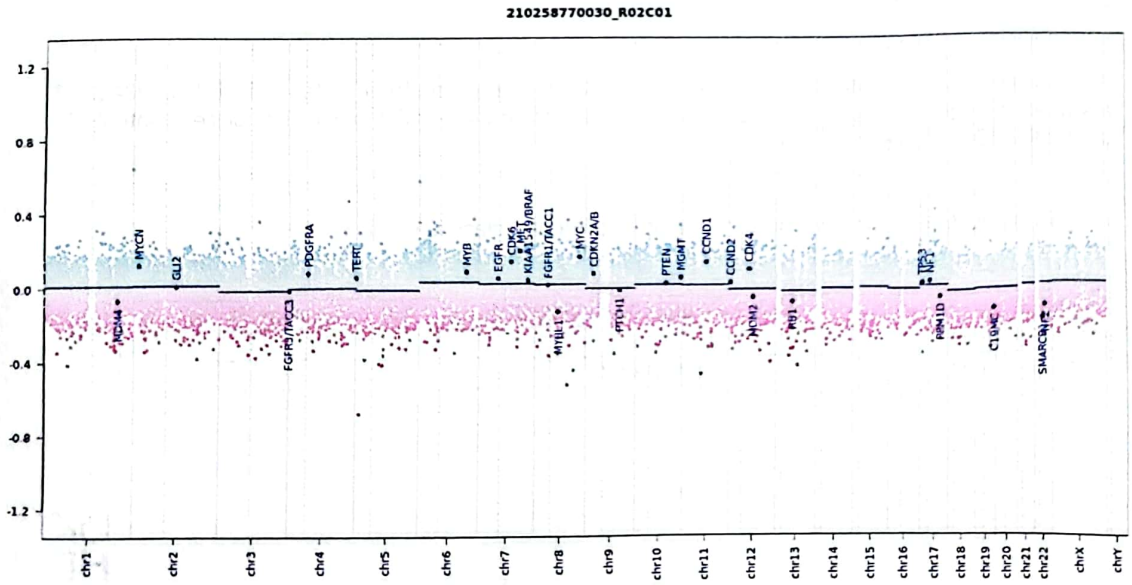
Superfamily			
	Family		
		Class	
			Subclass

*Only predictions with scores >= 0.9 are considered "classified". Scores < 0.9 could still be relevant for cases with low tumor content or poor DNA quality. Please exercise caution in interpreting such predictions.
Only the highest scores per level are shown

Description:
MC Pilocytic astrocytoma, midline: The "msc Supratentorial midline pilocytic astrocytoma" represents pilocytic astrocytoma and can also comprise tumors previously designated as pilomyxoid astrocytoma. They are located in supratentorial midline structures (thalamus, optic pathway, 3rd ventricle). These tumours typically carry MAPK pathway alterations, most commonly KIAA1549:BRAF fusion, but also some others (for example FGFR1 mutation is recurrently observed in this class and NF1-associated OPGs typically belong to this category). In case of high quality data the BRAF duplication can be appreciated on chromosome 7q (present in >60% of the reference cases). Besides this, most cases show a flat profile in copy number analysis.

Methylation Classification Sarcoma v12.3:
No score >= 0.3

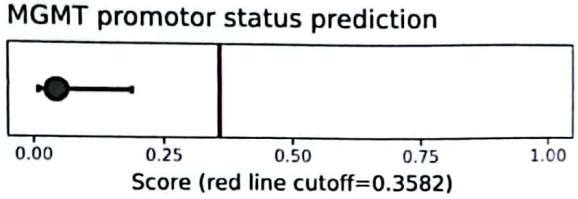
Methylation analyse - CNV:



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MGMT Promoter Methylation (MGMT-STP27):
Illumina Methylation Array:



Status	Estimate	CI lower	CI upper
unmethylated	0.043	0.0087	0.1877

Comment:
DNA methylation analysis is a recently developed diagnostic method that was evaluated in several studies (PMID 29539639, 28314689, 30786920, 31786093, 29967940) and which is recommended from international committees for brain tumor diagnostics (PMID 32307792, 32502305, 34185076). Prospective trials for generation of long-term data are ongoing. Studies also show added value for sarcoma diagnostics, which is currently undergoing validation (u.a. PMID 33047515, 33479225, 29881993, 32352245).

Molecular findings:
Methylation class "Pilocytic Astrocytoma, Midline"

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